

Deep Convolutional GAN

Aim:

Train a Deep convolutional GAN to generate realistic color images

Objective:

- ① implement DCGAN generator and discriminator architecture suitable for 32×32 color images
- ② Use a standard dataset of complex color images and preprocess images
- ③ Train with the stable DCGAN recipe and progressively monitor training
- ④ Print progress with epoch and loss
- ⑤ Plot for epoch vs loss

Pseudo Code:-① ~~LOAD~~ dataset

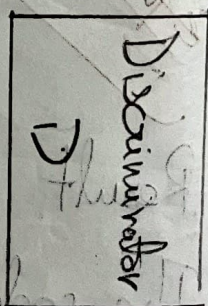
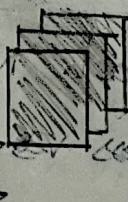
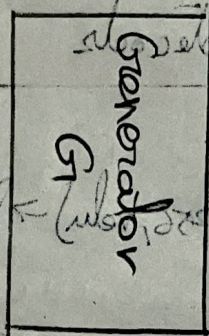
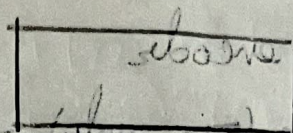
→ Load MNIST dataset (28×28 grayscale image)

② Define latent dimension

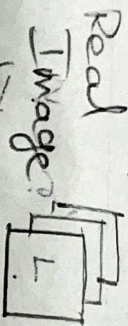
- set latent_dim = 2

Generative Adversarial Network (GANs)

Input



Yes/No



1. Define encoder
- Input size (width) $\rightarrow$ Output size (height) (e.g. 28x28 $\rightarrow$ 100 units)
2. Define decoder
- Input size (height) $\rightarrow$ Output size (width) (e.g. 100 units $\rightarrow$ 28x28)

3. Define AE (auto-encoder)
- Input size (width) $\rightarrow$ Output size (height) (e.g. 28x28 $\rightarrow$ 100 units $\rightarrow$ 28x28)
4. Plot loss vs. iteration and train weights
- Print output image and compare with real image

Adversarial process

Observation: L100 Loss: 1.1046 : G-Loss: 1.5642

Epoch 10 : D-Loss : 1.1046 : G-Loss : 1.5642

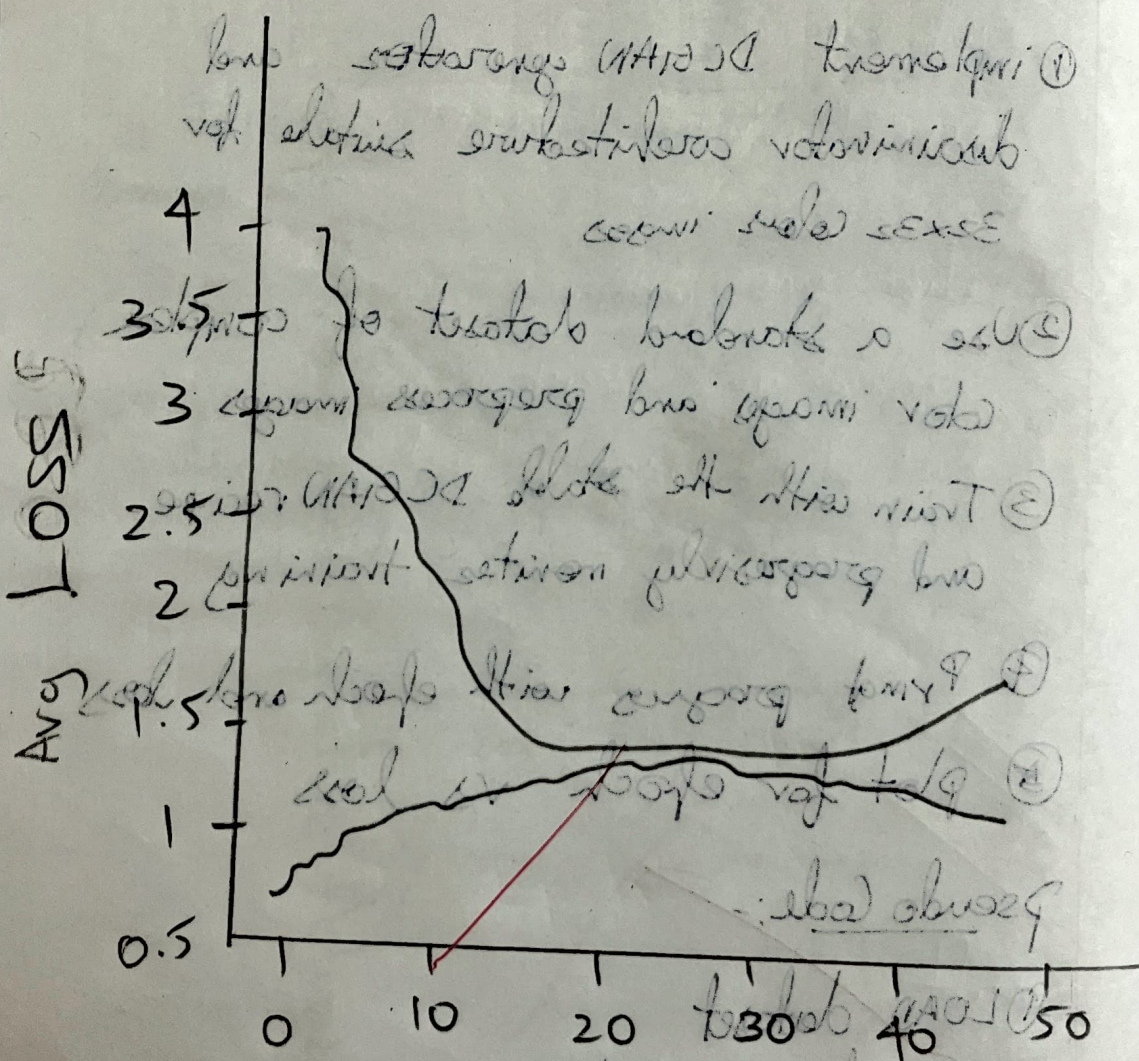
Epoch 20 : D-Loss : 1.1840 : G-Loss : 1.2321

Epoch 30 : D-Loss : 1.1772 : G-Loss : 1.2485

Epoch 40 : D-Loss : 1.0855 : G-Loss : 1.3522

Epoch 50 : D-Loss : 1.0116 : G-Loss : 1.4965

: 0.1046



(epoch 100) Loss: 1.0116 : G-Loss: 1.4965

Epoch

S = mib - test loss

③ define sampling layers

④ Build encoder

⑤ Build Decoder

⑥ Define VAE model

⑦ Define loss function

⑧ Training loop

for epoch = 1 to 50:

fit on training data

⑨ plot Loss

- plot loss vs epoch graph

⑩ END

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Result

The code executed successfully